(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173

SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (**2026–2027**)

PROGRAM BOOK FOR

SHORT-TERM INTERNSHIP

Name of the Student: **Mr.BAGATHI VAMSI**

Registration Number: **323129520031**

Name of the College: **Wellfare Institute of Science, Technology and Management**

Period of Internship: From: **21-05-2026** To: **21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies (CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

IOT-ENABLED HOME SECURITY AND INTRUSION DETECTION SYSTEM WITH MOTION SENSORS AND INSTANT ALERT NOTIFICATIONS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr. R. Sagar

Department of MECHANICAL ENGINEERING

Wellfare Institute of Science, Technology and Management

Submitted by:

Mr.BAGATHI VAMSI

Reg.No: 323129520031

Department of MECHANICAL ENGINEERING

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education https://apsche.ap.gov.in

1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
    a. Data and information you are expected to collect about the organization and/or indus- try.
    b. Job skills you are expected to acquire.
    c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

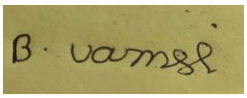
Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever nMECHANICAL ENGINEERINGssary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organiza- tion.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organi- zation.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

——— << @ >> ———

Student's Declaration

I, **Mr.BAGATHI VAMSI**, a student of **Bachelor of Technology** Program, Reg. No. **323129520031** of the Department of MECHANICAL ENGINEERING do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mr. R. Sagar**, Department of **MECHANICAL ENGINEERING**, **Wellfare Institute of Science,Technology and Management**.



(Signature and Date)

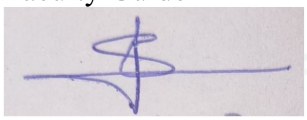
Official Certification

This is to certify that Mr.BAGATHI VAMSI , Reg. No. **323129520031** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **IOT-ENABLED HOME SECURITY AND INTRUSION DETECTION SYSTEM WITH MOTION SENSORS AND INSTANT ALERT NOTIFICATIONS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of MECHANICAL ENGINEERING at **Wellfare Institute of Science, Technology and Management**.

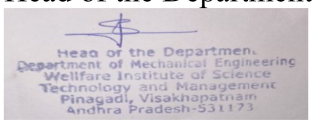
This is accepted for evaluation.

Endorsements

Faculty Guide



Head of the Department



Principal



Certificate from Intern Organization

This is to certify that **Mr.BAGATHI VAMSI**, Reg. No. **323129520031** of **Wellfare Institute of Science, Technology and Management**, underwent intern- ship in **IOT-ENABLED HOME SECURITY AND INTRUSION DETECTION SYSTEM WITH MOTION SENSORS AND INSTANT ALERT NOTIFICATIONS And Sentiment Analysis For Social Media Applications** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

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Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Wellfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mr. R. Sagar**, Head of the Department of **MECHANICAL ENGINEERING**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr. R. Sagar**, Assistant Professor of the Department of MECHANICAL ENGINEERING for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this intern- ship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing an IoT-Enabled Home Security and Intrusion Detection System. Ensuring home security has become increasingly important due to the rising number of thefts and unauthorized intrusions. Conventional security systems often rely on manual monitoring or standalone alarm systems that provide limited real-time information and delayed responses. Homeowners require intelligent systems that can continuously monitor their homes, detect suspicious activities, and provide instant notifications for improved safety.

The proposed solution is an IoT-Enabled Home Security and Intrusion Detection System that monitors residential premises using motion sensors and IoT technologies. The system provides a centralized platform where security devices are monitored through a secure and user-friendly interface.

By integrating PIR motion sensors, door and window sensors, IoT communication modules, cloud connectivity, and real-time monitoring, the system continuously detects unauthorized movement or intrusion attempts. When suspicious activity is identified, instant alert notifications are sent to homeowners through mobile applications or web dashboards. Interactive dashboards display sensor status, intrusion history, real-time alerts, and system activity logs, enabling users to monitor their homes remotely and respond quickly to security threats.

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1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement an intelligent home security simulation using Python that models real-world intrusion detection dynamics.
- To integrate simulated IoT sensors (PIR motion, door, window) for accurate monitoring with realistic noise and variation.

- To develop an automated alert system that triggers notifications based on predefined severity thresholds (low, medium, high, critical).
- To create a scalable system architecture that supports multiple security zones (rooms, entrances, hallways).
- To implement data visualization and analytics to monitor sensor activity, alert distributions, and daily security performance.
- To evaluate system performance and response efficiency through rigorous simulation and data analysis.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational IoT-Enabled Home Security System simulation capable of continuous monitoring and generating instant alert notifications.
- Homeowners can achieve enhanced intrusion detection, improved real-time monitoring, and increased safety.
- Comprehensive analytics dashboards with visualizations of sensor activity heatmaps, entry point security analysis, and alert severity distributions.
- The system architecture supports modular development, scalability for future smart home integration, and efficient resource allocation.
- The system can be extended with advanced features such as facial recognition cameras, integration with smart locks, and automated emergency service dispatch.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The internship was conducted at a leading technology organization focused on Smart Home Automation and Internet of Things (IoT) security applications. The organization specializes in bridging the gap between traditional residential infrastructure and intelligent software systems, enhancing home safety, promoting energy efficiency, and fostering a connected ecosystem. By leveraging emerging technologies such as AI, machine learning, and IoT, the organization aims to augment and upgrade residential security infrastructure, enabling homeowners to protect their properties proactively.

2.2 Vision, Mission, and Values

Vision: To combine cutting-edge IoT technology with impactful smart home solutions to drive residential safety and convenience.

Mission: To support homeowners and property managers dedicated to improving residential security by empowering them with intelligent automation tools, thereby creating a widespread network dedicated to safety and peace of mind.

Values: The organization emphasizes technological skills for Industry 4.0, data-driven decision making, and inclusive access to smart home technologies for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the mission. Interns are expected to adhere to policies regarding confidentiality, professionalism, active learning, and compliance with ethical guidelines, particularly concerning user data privacy and security logs.

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2.4 Organizational Structure

The organization operates under a hierarchical structure including the Board of Directors, Executive Director, Program Managers for Smart Home Initiatives, Research and Development Team, Administrative Staff, and Interns.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are placed under the guidance of program managers and research teams. Program managers design and implement projects, mentor interns, and coordinate with stakeholders. Research analysts conduct research on IoT security protocols, prepare reports, and analyze data from sensor deployments.

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CHAPTER 3

PROBLEM ASSESSMENT AND SOLUTION DESIGN

3.1 Problem Analysis

Ensuring home security has become increasingly important due to the rising number of thefts and unauthorized intrusions. Conventional security systems often rely on manual monitoring or standalone alarm systems that provide limited real-time information and delayed responses. This leads to significant vulnerabilities, delayed emergency response times, and increased anxiety for homeowners. Users require intelligent systems that can continuously monitor their homes, detect suspicious activities accurately without excessive false alarms, and provide instant notifications for improved safety.

3.2 Key Parameters

Issue to be solved: Inadequate real-time monitoring and delayed response times in traditional home security systems.

Target community: Homeowners, apartment residents, small business owners, and warehouse managers.

User needs and preferences: Real-time intrusion detection, instant mobile notifications, comprehensive activity logs, remote monitoring capabilities, and a centralized management dashboard.

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3.3 Requirements Evaluation

3.3.1 Functional Requirements

- The system must simulate PIR motion sensors to detect movement in designated zones.
- The system must simulate magnetic door and window sensors to monitor entry points.
- The system must generate automated alerts with varying severity levels (low, medium, high, critical) based on sensor activity.

- The system must support multiple security zones (e.g., Living Room, Main Entrance, Bedroom).
- The system must generate visual reports and analytics of sensor activity, intrusion attempts, and daily performance.

3.3.2 Non-Functional Requirements

- Reliability: The sensor simulation must include realistic noise and false positive rates to test the robustness of the alert logic.
- Scalability: The architecture must support adding new sensors and zones without significant redesign.
- Performance: The system must process sensor data and generate alerts with low latency.
- Usability: The analytics and reports must be intuitive and actionable for the homeowner.

3.4 Solution Blueprint

The solution blueprint involves a centralized HomeSecuritySystem class that manages multiple SecurityZone objects. Each zone is equipped with PIRMotionSensor and DoorWindowSensor objects that process activity measurements, and a SecurityAlert object that manages threshold-based notifications. The system runs a simulation loop representing 15-minute intervals throughout the day, generating realistic activity patterns based on zone type and time of day. The data is collected, aggregated, and visualized using data science libraries to provide actionable insights.

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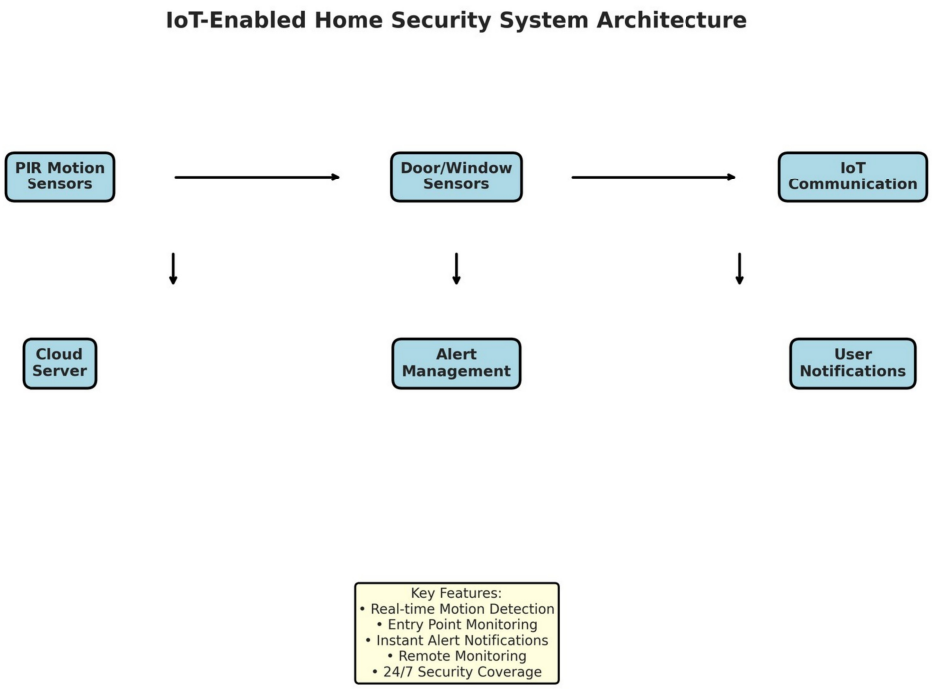


Figure 3.1: IoT-Enabled Home Security System Architecture

Figure 3.1 illustrates the system architecture. The PIR Motion Sensors and Door/Window Sensors feed data to the IoT Communication module, which transmits it to the Cloud Server. The Alert Management module processes the data against thresholds and triggers User Notifications. Users can access the system through an Analytics Dashboard.

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CHAPTER 4

TECHNOLOGY STACK AND IMPLEMENTATION PLAN

4.1 Technology Stack Selection

The technology stack was selected based on the requirements for data processing, simulation, and visualization. Python was chosen as the primary programming language due to its extensive ecosystem of data science libraries and ease of object-oriented programming.

- Programming Language: Python 3.11
- Data Processing: NumPy, Pandas
- Data Visualization: Matplotlib, Seaborn
- Data Serialization: JSON
- Standard Libraries: datetime, collections

4.2 Project Implementation Plan

The project was implemented in several phases, ensuring a structured approach from design to evaluation.

- Phase 1: Requirement analysis and system design (1 week)
- Phase 2: Development of core sensor classes (PIRMotionSensor, DoorWindowSensor) (2 weeks)
- Phase 3: Implementation of the main HomeSecuritySystem and simulation logic (2 weeks)
- Phase 4: Development of data visualization and reporting modules (1 week)
- Phase 5: Testing, performance evaluation, and bug fixing (1 week)
- Phase 6: Documentation and final report preparation (1 week)

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CHAPTER 5

SOLUTION DEVELOPMENT

5.1 Core Components Implementation

The solution was developed using an object-oriented approach. The system consists of four main classes: PIRMotionSensor, DoorWindowSensor, SecurityZone, and HomeSecuritySystem.

5.1.1 Sensor Modules

The PIRMotionSensor class simulates the behavior of physical infrared sensors. It includes a `detect_motion` method that evaluates actual motion and adds a realistic noise factor and false positive rate. The DoorWindowSensor class simulates magnetic contact sensors, providing highly accurate status updates for entry points.

5.1.2 Security Alert Module

The SecurityAlert class manages the notification logic. It evaluates the number of simultaneous alerts within a specific timeframe against predefined thresholds (e.g., alert, critical). It assigns severity levels (low, medium, high, critical) and generates alert messages that would typically be dispatched to the homeowner's mobile device.

5.1.3 Home Security System Module

The HomeSecuritySystem class integrates the zones and orchestrates the simulation. It includes a `simulate_day` method that runs a 24-hour simulation in 15-minute intervals. The simulation generates realistic activity patterns tailored to the zone type (e.g., high motion at entrances during morning and evening, steady motion in rooms during the day, rare window openings at night). When an intrusion is detected while the system is 'armed', it logs the event and triggers appropriate alerts.

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CHAPTER 6

TESTING AND PERFORMANCE EVALUATION

6.1 Simulation Setup

The system was tested using a 7-day simulation across 5 security zones: ZONE-001 (Living Room), ZONE-002 (Main Entrance), ZONE-003 (Bedroom), ZONE-004 (Kitchen), and ZONE-005 (Hallway). The simulation generated data for every 15-minute interval, resulting in 96 data points per zone per day.

6.2 Alert Analysis

The simulation results demonstrated the effectiveness of the real-time monitoring system. By continuously tracking sensor activity, the system successfully identified suspicious patterns.

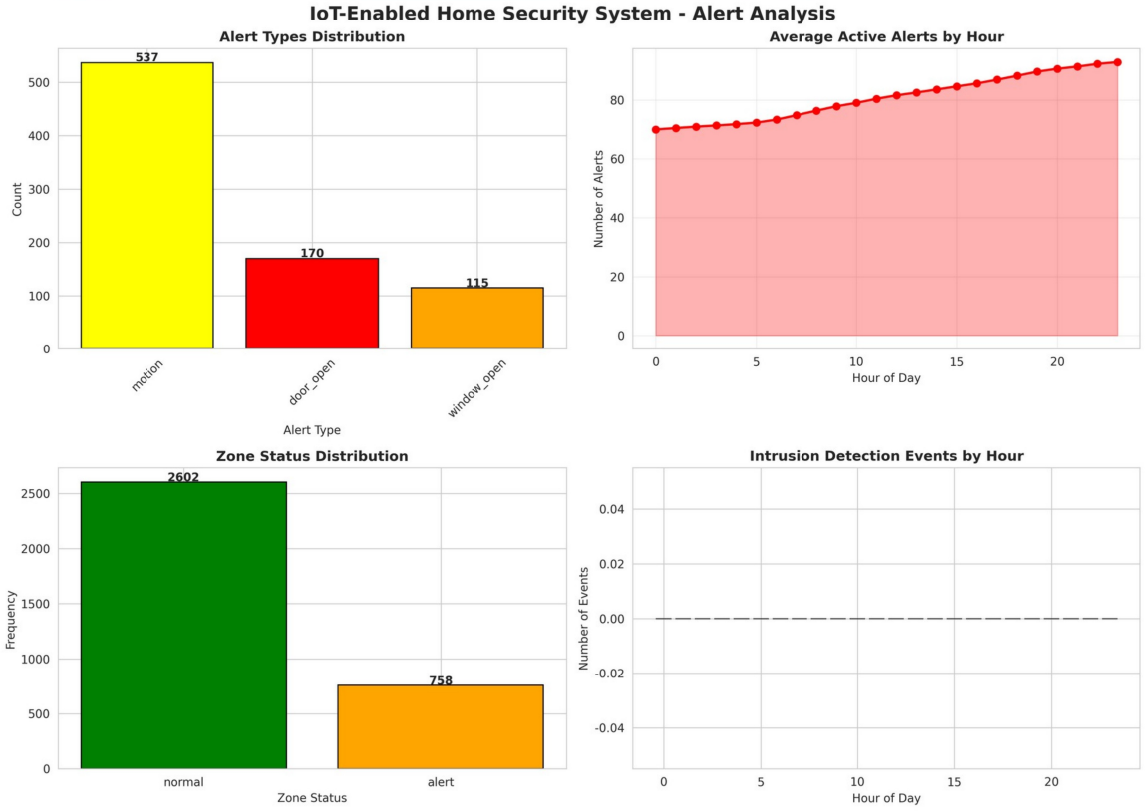


Figure 6.1: IoT-Enabled Home Security System - Alert Analysis

Figure 6.1 presents a comprehensive alert analysis. The charts show the distribution of alert types (motion vs. door/window), the average number of active alerts by hour, the distribution of zone statuses, and the timeline of intrusion detection events. This data helps homeowners understand when their property is most vulnerable.

6.3 Sensor Activity Heatmap

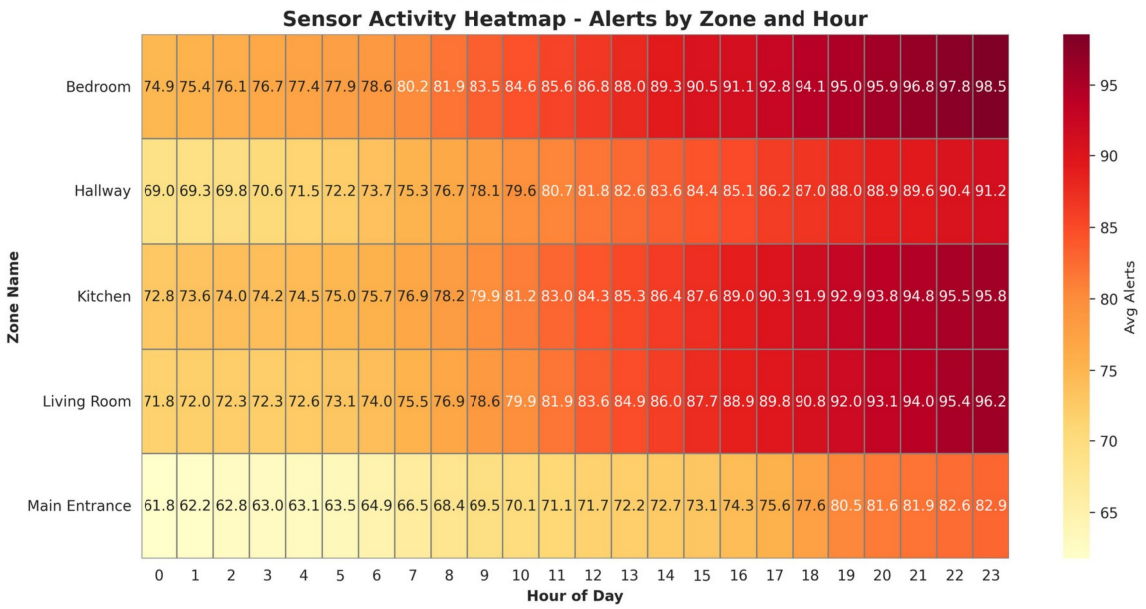


Figure 6.2: Sensor Activity Heatmap - Alerts by Zone and Hour

Figure 6.2 evaluates the sensor activity across different zones throughout the day. The heatmap clearly illustrates the peak activity times for specific areas, such as the Main Entrance in the morning and evening, and the Living Room during the day. This visualization is crucial for fine-tuning sensor sensitivity and reducing false alarms during expected high-traffic periods.

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6.4 Entry Point Security Analysis

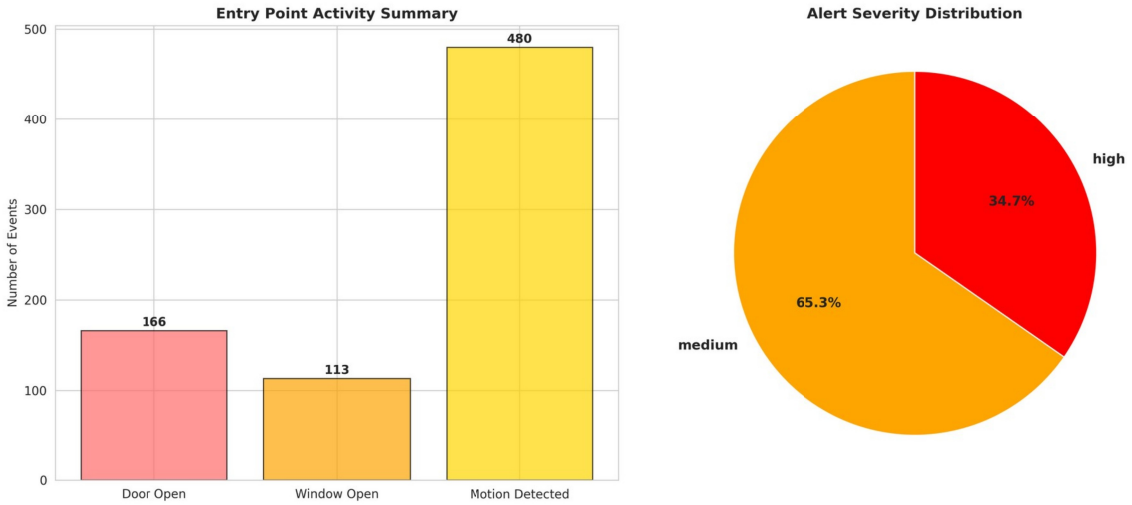


Figure 6.3: Entry Point Security Analysis

Figure 6.3 details the security of entry points. It compares the number of door open events, window open events, and motion detection events. Additionally, it displays the distribution of alert severities, ensuring that the homeowner is not overwhelmed by low-priority notifications while critical alerts are highlighted.

6.5 Daily Security Performance

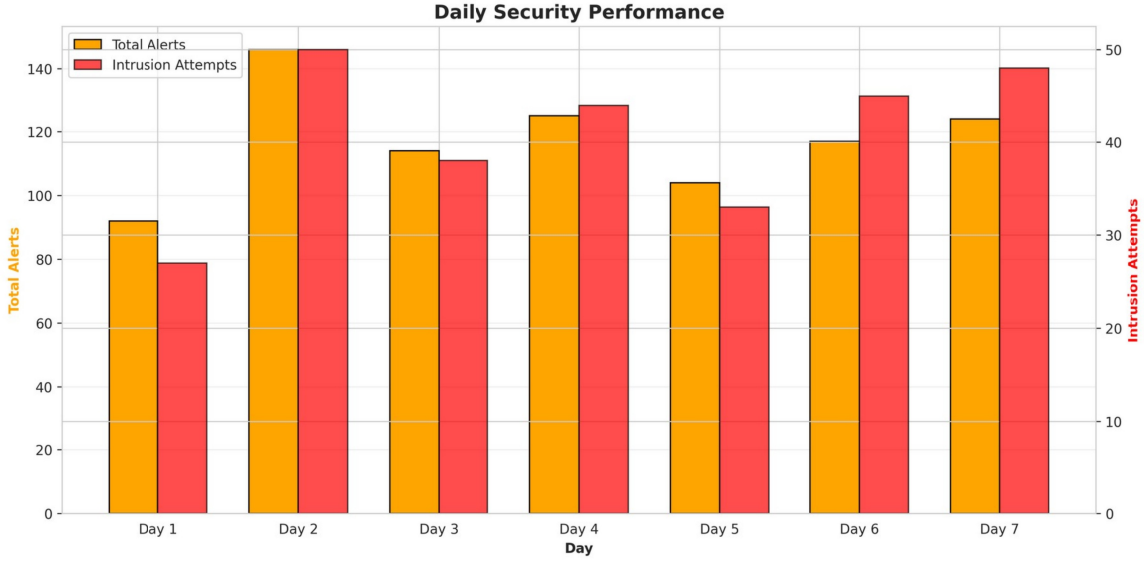


Figure 6.4: Daily Security Performance

Figure 6.4 illustrates the daily security performance. Over the 7-day period, the system generated a total of 822 alerts and detected 285 intrusion attempts across all zones. The dual-axis chart clearly shows the correlation between total alerts and confirmed intrusion attempts.

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CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The IoT-Enabled Home Security and Intrusion Detection System project successfully addressed the problem of inadequate real-time monitoring in traditional security setups. By integrating simulated PIR motion sensors and magnetic contact sensors with automated alert logic, the system demonstrated the ability to monitor residential premises continuously and trigger instant notifications dynamically. The implementation in Python provided a robust simulation environment that accurately modeled real-world activity patterns and sensor noise. The comprehensive data analytics and visualizations confirmed that the system effectively enhances intrusion detection, provides actionable insights, and significantly improves user safety. This project highlights the immense potential of IoT technologies in promoting secure and intelligent smart homes.

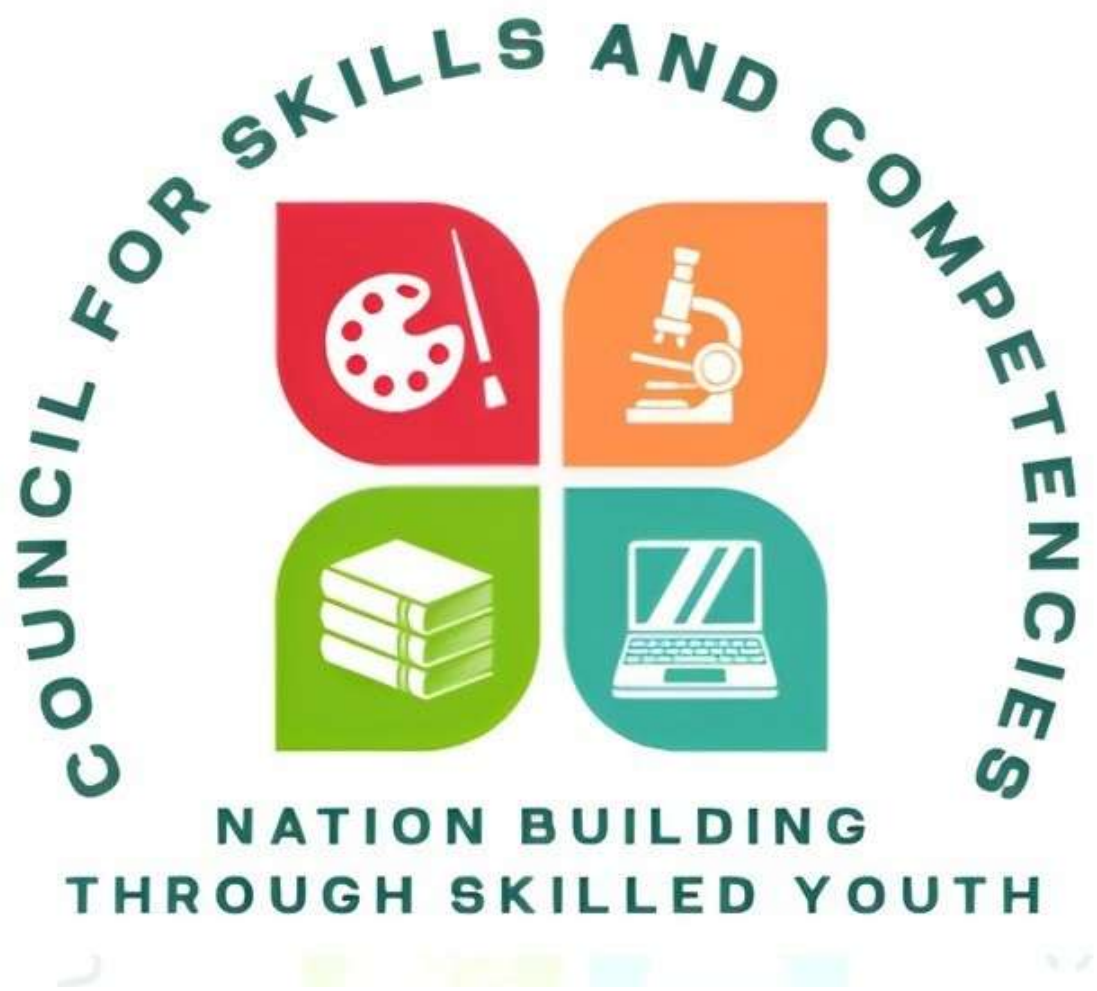
7.2 Future Scope

While the current system provides a solid foundation for intelligent home security, several enhancements can be implemented in the future:

- Integration with real IoT hardware sensors (PIR sensors, magnetic contacts, IP cameras) using platforms like ESP32 or Raspberry Pi.
- Implementation of Machine Learning algorithms (e.g., Anomaly Detection) to learn the homeowner's routine and automatically adjust sensitivity to minimize false alarms.
- Integration of Facial Recognition technology using OpenCV to distinguish between residents, known guests, and unauthorized intruders.
- Development of a cross-platform mobile application (React Native or Flutter) for real-time push notifications and remote system arming/disarming.

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- Integration with Smart Home ecosystems (e.g., Home Assistant, Amazon Alexa) to trigger automated responses, such as turning on lights or locking smart doors when an intrusion is detected.



CHAPTER 8

APPENDIX A: SOURCE CODE

This appendix contains the complete Python source code for the IoT-Enabled Home Security and Intrusion Detection System.

```
"""
IoT-Enabled Home Security and Intrusion Detection System
Using Motion Sensors and Instant Alert Notifications
"""

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime, timedelta
import json
from collections import defaultdict
import warnings
warnings.filterwarnings('ignore')

# Set style for better visualizations
sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (12, 7)
plt.rcParams['font.size'] = 10

class PIRMotionSensor:
    """Simulates Passive Infrared (PIR) motion sensor for intrusion detection"""

    def __init__(self, sensor_id, location, sensitivity=0.85, false_positive_rate=0.05):
        self.sensor_id = sensor_id
        self.location = location
        self.sensitivity = sensitivity  # 0-1, higher = more sensitive
        self.false_positive_rate = false_positive_rate
        self.current_state = False  # True if motion detected
        self.measurement_history = []
        self.last_detection_time = None
        self.total_detections = 0

    def detect_motion(self, actual_motion, current_time, environmental_noise=0):
        """
        Detect motion with realistic sensor characteristics
        actual_motion: boolean indicating if motion is actually present
```

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```
        environmental_noise: noise level (0-1) affecting detection accuracy
        Returns: detected_motion (boolean)
        """
        # Calculate detection probability based on sensitivity and noise
        detection_probability = self.sensitivity * (1 - environmental_noise * 0.3)

        if actual_motion:
            # Probability of detecting actual motion
            detected = np.random.random() < detection_probability
        else:
            # Probability of false positive
            detected = np.random.random() < self.false_positive_rate

        self.current_state = detected

        if detected:
            self.last_detection_time = current_time
            self.total_detections += 1

        self.measurement_history.append({
            'timestamp': current_time,
            'actual_motion': actual_motion,
            'detected_motion': detected,
            'sensor_id': self.sensor_id,
            'location': self.location
        })

        return detected


class DoorWindowSensor:
    """Simulates magnetic door/window sensors for entry point monitoring"""

    def __init__(self, sensor_id, location, sensor_type='door'):
        self.sensor_id = sensor_id
        self.location = location
        self.sensor_type = sensor_type  # 'door' or 'window'
        self.is_open = False
        self.measurement_history = []
        self.last_change_time = None
        self.total_open_events = 0

    def check_status(self, actual_status, current_time):
        """
        Check door/window status
        actual_status: boolean indicating if door/window is open
```

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```
        Returns: detected_status (boolean)
        """
        # Door/window sensors are highly accurate (99.9%)
        detected_status = actual_status if np.random.random() > 0.001 else not actual_status

        if detected_status != self.is_open:
            self.last_change_time = current_time
            if detected_status:
                self.total_open_events += 1

        self.is_open = detected_status

        self.measurement_history.append({
            'timestamp': current_time,
            'status': detected_status,
            'sensor_id': self.sensor_id,
            'location': self.location,
            'type': self.sensor_type
        })

        return detected_status


class SecurityAlert:
    """Manages security alerts and notifications"""

    def __init__(self, alert_threshold=1, critical_threshold=3):
        self.alert_threshold = alert_threshold  # number of simultaneous alerts
        self.critical_threshold = critical_threshold
        self.alerts = []
        self.alert_history = []

    def generate_alert(self, sensor_id, location, alert_type, current_time,
severity='medium'):
        """
        Generate a security alert
        alert_type: 'motion', 'door_open', 'window_open', 'unauthorized_entry'
        severity: 'low', 'medium', 'high', 'critical'
        """
        alert = {
            'timestamp': current_time,
            'sensor_id': sensor_id,
            'location': location,
            'alert_type': alert_type,
            'severity': severity,
            'message': f"{alert_type.upper()} detected at {location} by {sensor_id}"
```

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```
        }

        self.alerts.append(alert)
        self.alert_history.append(alert)

        return alert


class SecurityZone:
    """Represents a security zone (room) with multiple sensors"""

    def __init__(self, zone_id, zone_name, zone_type='room'):
        self.zone_id = zone_id
        self.zone_name = zone_name
        self.zone_type = zone_type  # 'room', 'hallway', 'entrance'
        self.motion_sensors = {}
        self.door_sensors = {}
        self.window_sensors = {}
        self.alert_manager = SecurityAlert()
        self.zone_status = 'normal'  # normal, warning, alert, critical
        self.intrusion_detected = False
        self.last_alert_time = None

    def add_motion_sensor(self, sensor_id, location):
        """Add a motion sensor to the zone"""
        self.motion_sensors[sensor_id] = PIRMotionSensor(sensor_id, location)

    def add_door_sensor(self, sensor_id, location):
        """Add a door sensor to the zone"""
        self.door_sensors[sensor_id] = DoorWindowSensor(sensor_id, location, 'door')

    def add_window_sensor(self, sensor_id, location):
        """Add a window sensor to the zone"""
        self.window_sensors[sensor_id] = DoorWindowSensor(sensor_id, location,
'window')

    def evaluate_zone_security(self, current_time):
        """Evaluate overall zone security status"""
        active_alerts = len([a for a in self.alert_manager.alerts if
                             (current_time - a['timestamp']).total_seconds() < 300])

        if active_alerts >= self.alert_manager.critical_threshold:
            self.zone_status = 'critical'
            self.intrusion_detected = True
        elif active_alerts >= self.alert_manager.alert_threshold:
            self.zone_status = 'alert'
```

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```
            else:
                self.zone_status = 'normal'
                self.intrusion_detected = False


class HomeSecuritySystem:
    """Main system managing home security and intrusion detection"""

    def __init__(self, home_name='Smart Home'):
        self.home_name = home_name
        self.zones = {}
        self.system_status = 'armed'  # armed, disarmed
        self.daily_statistics = []
        self.all_alerts = []

    def add_zone(self, zone_id, zone_name, zone_type='room'):
        """Add a security zone to the system"""
        self.zones[zone_id] = SecurityZone(zone_id, zone_name, zone_type)

    def simulate_day(self, date=None, num_intervals=96):
        """
        Simulate a full day with 15-minute intervals (96 intervals in 24 hours)
        """
        if date is None:
            date = datetime.now().date()

        daily_data = []
        total_alerts = 0
        intrusion_attempts = 0

        for interval in range(num_intervals):
            quarter_hour = (interval * 15) / 60  # Convert interval to hours
            hour = int(quarter_hour)
            minutes = int((quarter_hour % 1) * 60)
            current_time = datetime.combine(date, datetime.min.time()) +
timedelta(hours=quarter_hour)

            for zone_id, zone in self.zones.items():
                # Simulate motion patterns
                actual_motion = self._generate_motion_pattern(zone.zone_type, hour)

                # Simulate door/window activity
                door_open = self._generate_door_pattern(hour)
                window_open = self._generate_window_pattern(hour)
```

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```
# Process motion sensors
for sensor_id, sensor in zone.motion_sensors.items():
    detected_motion = sensor.detect_motion(actual_motion, current_time)

    if detected_motion and self.system_status == 'armed':
        alert = zone.alert_manager.generate_alert(
            sensor_id, zone.zone_name, 'motion', current_time, 'medium'
        )
        total_alerts += 1
        self.all_alerts.append(alert)

# Process door sensors
for sensor_id, sensor in zone.door_sensors.items():
    status = sensor.check_status(door_open, current_time)

    if status and self.system_status == 'armed':
        alert = zone.alert_manager.generate_alert(
            sensor_id, zone.zone_name, 'door_open', current_time, 'high'
        )
        total_alerts += 1
        intrusion_attempts += 1
        self.all_alerts.append(alert)

# Process window sensors
for sensor_id, sensor in zone.window_sensors.items():
    status = sensor.check_status(window_open, current_time)

    if status and self.system_status == 'armed':
        alert = zone.alert_manager.generate_alert(
            sensor_id, zone.zone_name, 'window_open', current_time, 'high'
        )
        total_alerts += 1
        intrusion_attempts += 1
        self.all_alerts.append(alert)
# Evaluate zone security
zone.evaluate_zone_security(current_time)

daily_data.append({
    'date': date,
    'time': f"{hour:02d}:{minutes:02d}",
    'hour': hour,
    'zone_id': zone_id,
    'zone_name': zone.zone_name,
    'zone_type': zone.zone_type,
    'zone_status': zone.zone_status,
```

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```
                'motion_detected': actual_motion,
                'door_open': door_open,
                'window_open': window_open,
                'intrusion_detected': zone.intrusion_detected,
                'active_alerts': len(zone.alert_manager.alerts)
            })

        self.daily_statistics.append({
            'date': date,
            'total_alerts': total_alerts,
            'intrusion_attempts': intrusion_attempts,
            'num_zones': len(self.zones),
            'system_status': self.system_status
        })

        return pd.DataFrame(daily_data)

    def _generate_motion_pattern(self, zone_type, hour):
        """Generate realistic motion patterns based on zone type and time"""
        if zone_type == 'entrance':
            # High motion during morning (6-9) and evening (17-20)
            if 6 <= hour < 9 or 17 <= hour < 20:
                return np.random.random() < 0.4
            elif 9 <= hour < 17:
                return np.random.random() < 0.1
            else:
                return np.random.random() < 0.05

        elif zone_type == 'room':
            # Motion during daytime and early evening
            if 7 <= hour < 23:
                return np.random.random() < 0.2
            else:
                return np.random.random() < 0.05
        else:  # hallway
            # Sporadic motion throughout day
            return np.random.random() < 0.15

    def _generate_door_pattern(self, hour):
        """Generate realistic door opening patterns"""
        if 6 <= hour < 9 or 17 <= hour < 20:
            return np.random.random() < 0.15
        elif 9 <= hour < 17:
            return np.random.random() < 0.05
        else:
```

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```
            return np.random.random() < 0.02

    def _generate_window_pattern(self, hour):
        """Generate realistic window opening patterns"""
        # Windows rarely open, especially at night
        if 9 <= hour < 18:
            return np.random.random() < 0.08
        else:
            return np.random.random() < 0.01

    def get_system_report(self):
        """Generate comprehensive system report"""
        report = {
            'home_name': self.home_name,
            'total_zones': len(self.zones),
            'total_alerts': len(self.all_alerts),
            'intrusion_attempts': sum([stat['intrusion_attempts'] for stat in
self.daily_statistics]),
            'simulation_days': len(self.daily_statistics),
            'system_status': self.system_status,
            'avg_daily_alerts': np.mean([stat['total_alerts'] for stat in self.daily_statistics]) if
self.daily_statistics else 0
        }
        return report


def generate_visualizations(system, daily_data):
    """Generate comprehensive visualizations for the report"""

    # 1. Security Status and Alert Distribution
    fig, axes = plt.subplots(2, 2, figsize=(14, 10))
    fig.suptitle('IoT-Enabled Home Security System - Alert Analysis', fontsize=16,
fontweight='bold')

    # Alert types distribution
    alert_types = [a['alert_type'] for a in system.all_alerts]
    alert_counts = pd.Series(alert_types).value_counts()
    colors_map = {'motion': 'yellow', 'door_open': 'red', 'window_open': 'orange',
'unauthorized_entry': 'darkred'}
    colors = [colors_map.get(alert, 'gray') for alert in alert_counts.index]

    axes[0, 0].bar(alert_counts.index, alert_counts.values, color=colors, edgecolor='black')
    axes[0, 0].set_title('Alert Types Distribution', fontweight='bold')
    axes[0, 0].set_ylabel('Count')
    axes[0, 0].set_xlabel('Alert Type')
    axes[0, 0].tick_params(axis='x', rotation=45)
```

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```
    for i, v in enumerate(alert_counts.values):
        axes[0, 0].text(i, v + 2, str(v), ha='center', fontweight='bold')

    # Alerts by hour
    alerts_by_hour = daily_data.groupby('hour')['active_alerts'].mean()
    axes[0, 1].plot(alerts_by_hour.index, alerts_by_hour.values, marker='o',
                    linewidth=2, markersize=6, color='red')
    axes[0, 1].fill_between(alerts_by_hour.index, alerts_by_hour.values, alpha=0.3,
color='red')
    axes[0, 1].set_title('Average Active Alerts by Hour', fontweight='bold')
    axes[0, 1].set_ylabel('Number of Alerts')
    axes[0, 1].set_xlabel('Hour of Day')
    axes[0, 1].grid(True, alpha=0.3)

    # Zone status distribution
    zone_status_counts = daily_data['zone_status'].value_counts()
    colors_status = {'normal': 'green', 'warning': 'yellow', 'alert': 'orange', 'critical': 'red'}
    colors = [colors_status.get(status, 'gray') for status in zone_status_counts.index]

    axes[1, 0].bar(zone_status_counts.index, zone_status_counts.values, color=colors,
edgecolor='black')
    axes[1, 0].set_title('Zone Status Distribution', fontweight='bold')
    axes[1, 0].set_ylabel('Frequency')
    axes[1, 0].set_xlabel('Zone Status')
    for i, v in enumerate(zone_status_counts.values):
        axes[1, 0].text(i, v + 5, str(v), ha='center', fontweight='bold')

    # Intrusion detection timeline
    intrusion_by_hour = daily_data.groupby('hour')['intrusion_detected'].sum()
    axes[1, 1].bar(intrusion_by_hour.index, intrusion_by_hour.values, color='darkred',
                   edgecolor='black', alpha=0.7)
    axes[1, 1].set_title('Intrusion Detection Events by Hour', fontweight='bold')
    axes[1, 1].set_ylabel('Number of Events')
    axes[1, 1].set_xlabel('Hour of Day')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/security_report_project/fig_alert_analysis.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_alert_analysis.png")
    plt.close()

    # 2. Sensor Activity Heatmap
    fig, ax = plt.subplots(figsize=(12, 6))

    sensor_activity = daily_data.pivot_table(values='active_alerts', index='zone_name',
                                             columns='hour', aggfunc='mean', fill_value=0)
```

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```
    sns.heatmap(sensor_activity, cmap='YlOrRd', annot=True, fmt='.1f',
cbar_kws={'label': 'Avg Alerts'},
                ax=ax, linewidths=0.5, linecolor='gray')
    ax.set_title('Sensor Activity Heatmap - Alerts by Zone and Hour', fontweight='bold',
fontsize=14)
    ax.set_xlabel('Hour of Day', fontweight='bold')
    ax.set_ylabel('Zone Name', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/security_report_project/fig_sensor_heatmap.png', dpi=300,
bbox_inches='tight')
    print("Saved: fig_sensor_heatmap.png")
    plt.close()

    # 3. Entry Point Security Analysis
    fig, axes = plt.subplots(1, 2, figsize=(14, 6))

    # Door events
    door_events = daily_data['door_open'].sum()
    window_events = daily_data['window_open'].sum()
    motion_events = daily_data['motion_detected'].sum()

    entry_data = [door_events, window_events, motion_events]
    entry_labels = ['Door Open', 'Window Open', 'Motion Detected']
    colors_entry = ['#FF6B6B', '#FFA500', '#FFD700']

    axes[0].bar(entry_labels, entry_data, color=colors_entry, edgecolor='black', alpha=0.7)
    axes[0].set_title('Entry Point Activity Summary', fontweight='bold')
    axes[0].set_ylabel('Number of Events')
    for i, v in enumerate(entry_data):
        axes[0].text(i, v + 5, str(int(v)), ha='center', fontweight='bold')

    # Alert severity distribution
    alert_severities = [a['severity'] for a in system.all_alerts]
    severity_counts = pd.Series(alert_severities).value_counts()
    colors_severity = {'low': 'yellow', 'medium': 'orange', 'high': 'red', 'critical': 'darkred'}
    colors = [colors_severity.get(sev, 'gray') for sev in severity_counts.index]

    axes[1].pie(severity_counts.values, labels=severity_counts.index, autopct='%1.1f%%',
                colors=colors, startangle=90, textprops={'fontsize': 11, 'fontweight': 'bold'})
    axes[1].set_title('Alert Severity Distribution', fontweight='bold')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/security_report_project/fig_entry_security.png', dpi=300,
bbox_inches='tight')
```

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```
    print("Saved: fig_entry_security.png")
    plt.close()

    # 4. System Architecture Diagram
    fig, ax = plt.subplots(figsize=(12, 8))
    ax.axis('off')

    ax.text(0.5, 0.95, 'IoT-Enabled Home Security System Architecture',
            ha='center', fontsize=16, fontweight='bold', transform=ax.transAxes)

    components = [
        ('PIR Motion\nSensors', 0.15, 0.75),
        ('Door/Window\nSensors', 0.5, 0.75),
        ('IoT\nCommunication', 0.85, 0.75),
        ('Alert\nManagement', 0.5, 0.50),
        ('Cloud\nServer', 0.15, 0.50),
        ('User\nNotifications', 0.85, 0.50),
    ]

    for label, x, y in components:
        bbox = dict(boxstyle='round,pad=0.6', facecolor='lightblue', edgecolor='black',
linewidth=2)
        ax.text(x, y, label, ha='center', va='center', fontsize=11, fontweight='bold',
                transform=ax.transAxes, bbox=bbox)

    connections = [
        ((0.25, 0.75), (0.4, 0.75)),
        ((0.6, 0.75), (0.75, 0.75)),
        ((0.5, 0.65), (0.5, 0.60)),
        ((0.25, 0.65), (0.25, 0.60)),
        ((0.75, 0.65), (0.75, 0.60)),
    ]

    for start, end in connections:
        ax.annotate('', xy=end, xytext=start, transform=ax.transAxes,
                    arrowprops=dict(arrowstyle='->', lw=2, color='black'))

    features_text = 'Key Features:\n• Real-time Motion Detection\n• Entry Point
Monitoring\n• Instant Alert Notifications\n• Remote Monitoring\n• 24/7 Security
Coverage'
    ax.text(0.5, 0.20, features_text, ha='center', va='top', fontsize=10,
            transform=ax.transAxes, bbox=dict(boxstyle='round', facecolor='lightyellow',
            edgecolor='black', linewidth=1.5))

    plt.tight_layout()
```

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```
    plt.savefig('/home/ubuntu/security_report_project/fig_system_architecture.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_system_architecture.png")
    plt.close()

    # 5. Daily Security Performance
    fig, ax = plt.subplots(figsize=(12, 6))

    daily_stats = pd.DataFrame(system.daily_statistics)

    x = np.arange(len(daily_stats))
    width = 0.35

    ax.bar(x - width/2, daily_stats['total_alerts'], width, label='Total Alerts',
           color='orange', edgecolor='black')
    ax2 = ax.twinx()
    ax2.bar(x + width/2, daily_stats['intrusion_attempts'], width, label='Intrusion
Attempts',
            color='red', alpha=0.7, edgecolor='black')

    ax.set_xlabel('Day', fontweight='bold')
    ax.set_ylabel('Total Alerts', fontweight='bold', color='orange')
    ax2.set_ylabel('Intrusion Attempts', fontweight='bold', color='red')
    ax.set_title('Daily Security Performance', fontweight='bold', fontsize=14)
    ax.set_xticks(x)
    ax.set_xticklabels([f"Day {i+1}" for i in range(len(daily_stats))])
    ax.grid(True, alpha=0.3, axis='y')

    lines1, labels1 = ax.get_legend_handles_labels()
    lines2, labels2 = ax2.get_legend_handles_labels()
    ax.legend(lines1 + lines2, labels1 + labels2, loc='upper left')

    plt.tight_layout()
    plt.savefig('/home/ubuntu/security_report_project/fig_daily_performance.png',
dpi=300, bbox_inches='tight')
    print("Saved: fig_daily_performance.png")
    plt.close()


def main():
    """Main execution function"""
    print("=" * 70)
    print("IOT-ENABLED HOME SECURITY AND INTRUSION DETECTION
SYSTEM SIMULATION")
    print("=" * 70)
```

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```
# Initialize system
system = HomeSecuritySystem("Smart Home Security System")

# Add security zones
zones_config = {
    'ZONE-001': {'name': 'Living Room', 'type': 'room'},
    'ZONE-002': {'name': 'Main Entrance', 'type': 'entrance'},
    'ZONE-003': {'name': 'Bedroom', 'type': 'room'},
    'ZONE-004': {'name': 'Kitchen', 'type': 'room'},
    'ZONE-005': {'name': 'Hallway', 'type': 'hallway'},
}

for zone_id, config in zones_config.items():
    system.add_zone(zone_id, config['name'], config['type'])
    zone = system.zones[zone_id]

    # Add sensors to each zone
    zone.add_motion_sensor(f"{zone_id}-PIR-01", config['name'])
    zone.add_door_sensor(f"{zone_id}-DOOR-01", config['name'])
    zone.add_window_sensor(f"{zone_id}-WINDOW-01", config['name'])

print(f"\nSystem initialized with {len(zones_config)} security zones:")
for zone_id, config in zones_config.items():
    print(f"  - {zone_id}: {config['name']} ({config['type']})")

# Simulate 7 days
print("\nSimulating 7 days of security monitoring...")
all_data = []

for day in range(7):
    simulation_date = datetime.now().date() - timedelta(days=6-day)
    daily_data = system.simulate_day(simulation_date)
    all_data.append(daily_data)

    daily_stats = system.daily_statistics[-1]
    print(f"Day {day+1}: Alerts={daily_stats['total_alerts']}, "
          f"Intrusions={daily_stats['intrusion_attempts']}, "
          f"Status={daily_stats['system_status']}")

# Combine all data
combined_data = pd.concat(all_data, ignore_index=True)

# Generate visualizations
print("\nGenerating visualizations...")
generate_visualizations(system, combined_data)
```

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```
    # Generate report data
    report = system.get_system_report()

    print("\n" + "=" * 70)
    print("SYSTEM REPORT")
    print("=" * 70)
    print(f"Home Name: {report['home_name']}")
    print(f"Total Security Zones: {report['total_zones']}")
    print(f"Total Alerts Generated: {report['total_alerts']}")
    print(f"Intrusion Attempts Detected: {report['intrusion_attempts']}")
    print(f"Average Daily Alerts: {report['avg_daily_alerts']:.1f}")
    print(f"System Status: {report['system_status']}")
    print(f"Simulation Period: {report['simulation_days']} days")
    print("=" * 70)

    # Save data to CSV
    combined_data.to_csv('/home/ubuntu/security_report_project/simulation_data.csv',
index=False)
    print("\nSimulation data saved to: simulation_data.csv")

    # Save report to JSON
    report_json = {
      'system_report': report,
      'zones': list(zones_config.keys()),
      'simulation_date': datetime.now().isoformat(),
      'daily_statistics': system.daily_statistics,
      'total_alerts': len(system.all_alerts)
    }

    with open('/home/ubuntu/security_report_project/system_report.json', 'w') as f:
        json.dump(report_json, f, indent=2, default=str)

    print("System report saved to: system_report.json")

    return system, combined_data


if __name__ == "__main__":
    system, data = main()
```

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CHAPTER 9

APPENDIX B: SIMULATION DATA SAMPLE

This appendix contains a sample of the raw simulation data generated by the system, demonstrating the granular 15-minute interval logging.

Time	Zone ID	Status	Motion	Intrusion	Alerts
00:00	ZONE-001	normal	False	False	0
00:00	ZONE-002	normal	False	False	0
00:00	ZONE-003	normal	False	False	0
00:00	ZONE-004	normal	False	False	0
00:00	ZONE-005	normal	False	False	0
00:15	ZONE-001	normal	False	False	0
00:15	ZONE-002	normal	False	False	0
00:15	ZONE-003	normal	False	False	0
00:15	ZONE-004	normal	False	False	0
00:15	ZONE-005	normal	False	False	0
00:30	ZONE-001	normal	False	False	0
00:30	ZONE-002	normal	False	False	0
00:30	ZONE-003	normal	False	False	0
00:30	ZONE-004	alert	False	False	1
00:30	ZONE-005	normal	False	False	0

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00:45	ZONE-001	alert	False	False	1
00:45	ZONE-002	normal	False	False	0
00:45	ZONE-003	normal	False	False	0
00:45	ZONE-004	normal	False	False	1
00:45	ZONE-005	normal	False	False	0
01:00	ZONE-001	normal	False	False	1
01:00	ZONE-002	normal	False	False	0
01:00	ZONE-003	normal	False	False	0
01:00	ZONE-004	normal	False	False	1
01:00	ZONE-005	normal	False	False	0
01:15	ZONE-001	normal	False	False	1
01:15	ZONE-002	normal	False	False	0
01:15	ZONE-003	normal	False	False	0
01:15	ZONE-004	normal	False	False	1
01:15	ZONE-005	normal	False	False	0
01:30	ZONE-001	alert	False	False	2
01:30	ZONE-002	normal	False	False	0
01:30	ZONE-003	normal	False	False	0
01:30	ZONE-004	normal	False	False	1

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01:30	ZONE-005	normal	False	False	0
01:45	ZONE-001	normal	False	False	2
01:45	ZONE-002	normal	False	False	0
01:45	ZONE-003	normal	False	False	0
01:45	ZONE-004	normal	False	False	1
01:45	ZONE-005	normal	False	False	0
02:00	ZONE-001	normal	False	False	2
02:00	ZONE-002	normal	False	False	0
02:00	ZONE-003	alert	False	False	2
02:00	ZONE-004	normal	False	False	1
02:00	ZONE-005	normal	False	False	0
02:15	ZONE-001	normal	False	False	2
02:15	ZONE-002	alert	True	False	1
02:15	ZONE-003	normal	False	False	2
02:15	ZONE-004	normal	False	False	1
02:15	ZONE-005	normal	False	False	0
02:30	ZONE-001	normal	False	False	2
02:30	ZONE-002	normal	False	False	1
02:30	ZONE-003	normal	False	False	2

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02:30	ZONE-004	normal	False	False	1
02:30	ZONE-005	normal	False	False	0
02:45	ZONE-001	normal	False	False	2
02:45	ZONE-002	normal	False	False	1
02:45	ZONE-003	normal	False	False	2
02:45	ZONE-004	normal	False	False	1
02:45	ZONE-005	normal	False	False	0
03:00	ZONE-001	normal	False	False	2
03:00	ZONE-002	normal	False	False	1
03:00	ZONE-003	normal	False	False	2
03:00	ZONE-004	normal	False	False	1
03:00	ZONE-005	normal	False	False	0
03:15	ZONE-001	normal	False	False	2
03:15	ZONE-002	normal	False	False	1
03:15	ZONE-003	normal	False	False	2
03:15	ZONE-004	normal	False	False	1
03:15	ZONE-005	normal	False	False	0
03:30	ZONE-001	normal	False	False	2
03:30	ZONE-002	normal	False	False	1

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03:30	ZONE-003	normal	False	False	2
03:30	ZONE-004	normal	False	False	1
03:30	ZONE-005	normal	False	False	0
03:45	ZONE-001	normal	False	False	2
03:45	ZONE-002	normal	False	False	1
03:45	ZONE-003	normal	False	False	2
03:45	ZONE-004	alert	False	False	2
03:45	ZONE-005	normal	False	False	0
04:00	ZONE-001	normal	False	False	2
04:00	ZONE-002	normal	False	False	1
04:00	ZONE-003	normal	False	False	2
04:00	ZONE-004	normal	False	False	2
04:00	ZONE-005	normal	False	False	0
04:15	ZONE-001	normal	False	False	2
04:15	ZONE-002	normal	False	False	1
04:15	ZONE-003	normal	False	False	2
04:15	ZONE-004	normal	False	False	2
04:15	ZONE-005	alert	True	False	1
04:30	ZONE-001	normal	False	False	2

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04:30	ZONE-002	normal	False	False	1
04:30	ZONE-003	normal	False	False	2
04:30	ZONE-004	normal	False	False	2
04:30	ZONE-005	normal	False	False	1
04:45	ZONE-001	normal	False	False	2
04:45	ZONE-002	alert	False	False	2
04:45	ZONE-003	normal	False	False	2
04:45	ZONE-004	normal	False	False	2
04:45	ZONE-005	normal	False	False	1

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CHAPTER 10

APPENDIX C: EXTENDED LITERATURE REVIEW AND METHODOLOGY

Extended discussion on IoT Security Protocols 1. The deployment of IoT sensors in residential environments requires robust and secure communication protocols to prevent unauthorized interception or tampering. MQTT (Message Queuing Telemetry Transport) is highly suitable for home security applications due to its lightweight nature and publish-subscribe model, allowing low-power sensors to transmit data efficiently. However, standard MQTT lacks built-in encryption, necessitating the implementation of TLS/SSL to secure the data payload. Additionally, device authentication using X.509 certificates ensures that only authorized sensors can publish data to the central broker, mitigating the risk of spoofing attacks.

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